CSE-251
Electronic Circuits

Lecture-02

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Today's Overview:

- *Electricity*
- *Types of Electricity*
- *Atoms*
- *Electrical Units and Symbols*
- *Multiples and Sub-multiples*
- *Electrical and Electronic Devices*
- *Importance of Electronics*

Electricity

Electricity is a fundamental form of energy observable in positive and negative forms that occurs naturally (as in lightning) or is produced (as in a generator) and that is expressed in terms of the movement and interaction of electrons.

Electricity is a basic part of nature and it is one of our most widely used forms of energy. Before electricity generation began over 100 years ago, houses were lit with kerosene lamps, food was cooled in iceboxes, and rooms were warmed by wood-burning or coal-burning stoves.

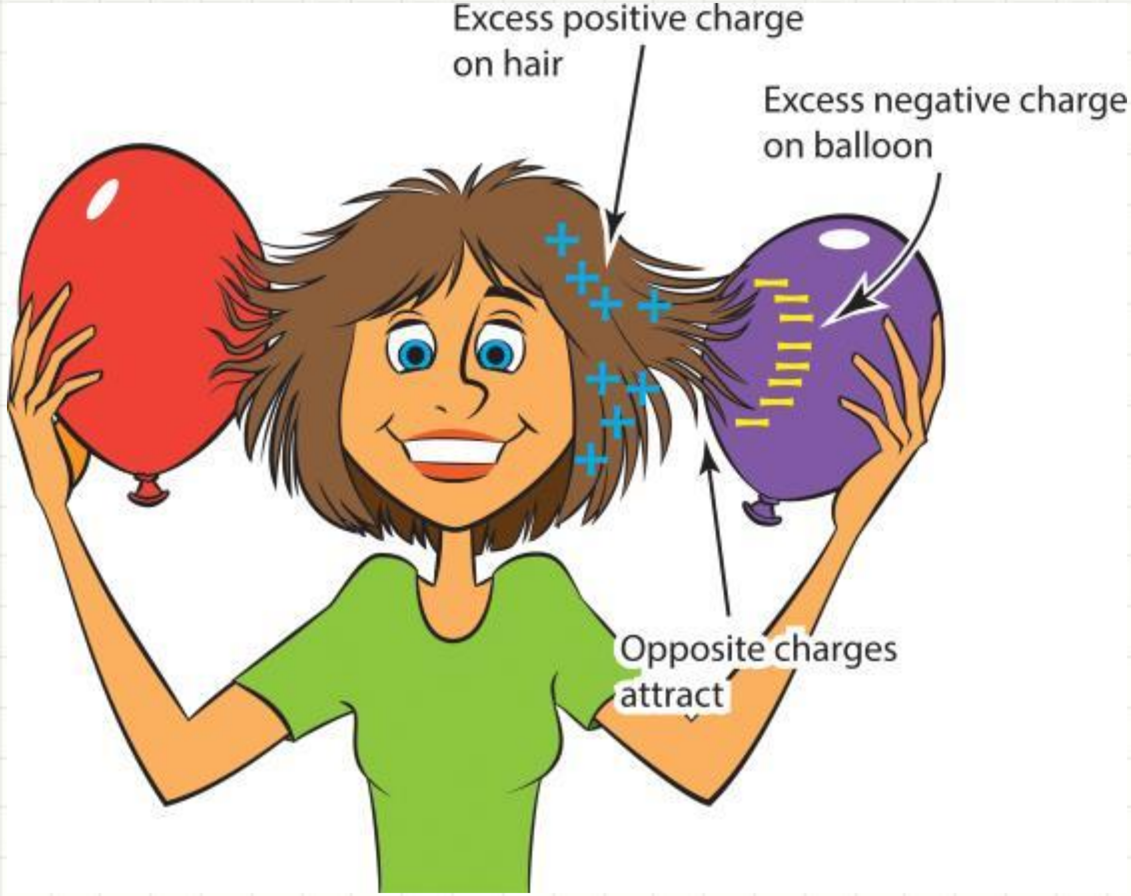
At 1879, Thomas Edison helped change everyone's life, he perfected his invention, the electric light bulb.

In the late-1800s, Nikola Tesla pioneered the generation, transmission, and use of alternating current (AC) electricity, which can be transmitted over much greater distances than direct current.

Types of Electricity

There are mainly two types of electricity:

- Static Electricity
- Dynamic / Current Electricity



Static Electricity:

Static electricity is the result of an imbalance between negative and positive charges in an object.

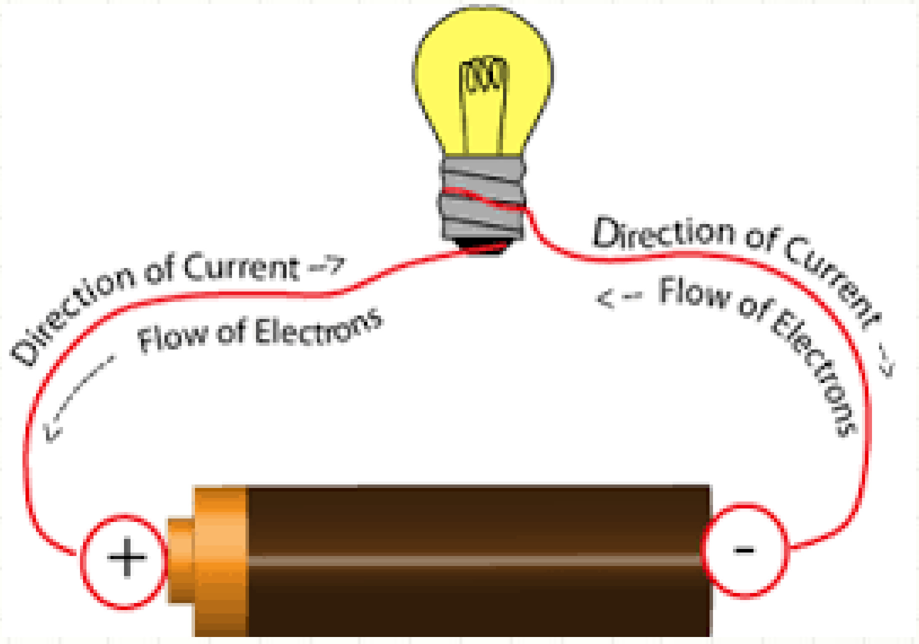
These charges can build up on the surface of an object until they find a way to be released or discharged.

Current Electricity:

Current electricity is defined as the flow of electrons from one section of the circuit to another.

There are two types of current electricity as follows:

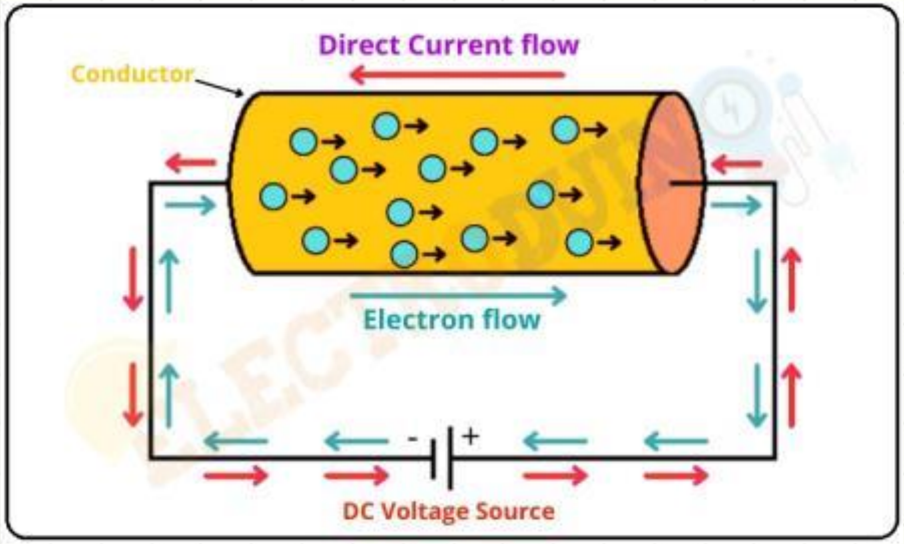
- Direct Current (DC)
- Alternating Current (AC)



Direct Current:

The current electricity whose direction remains the same is known as direct current.

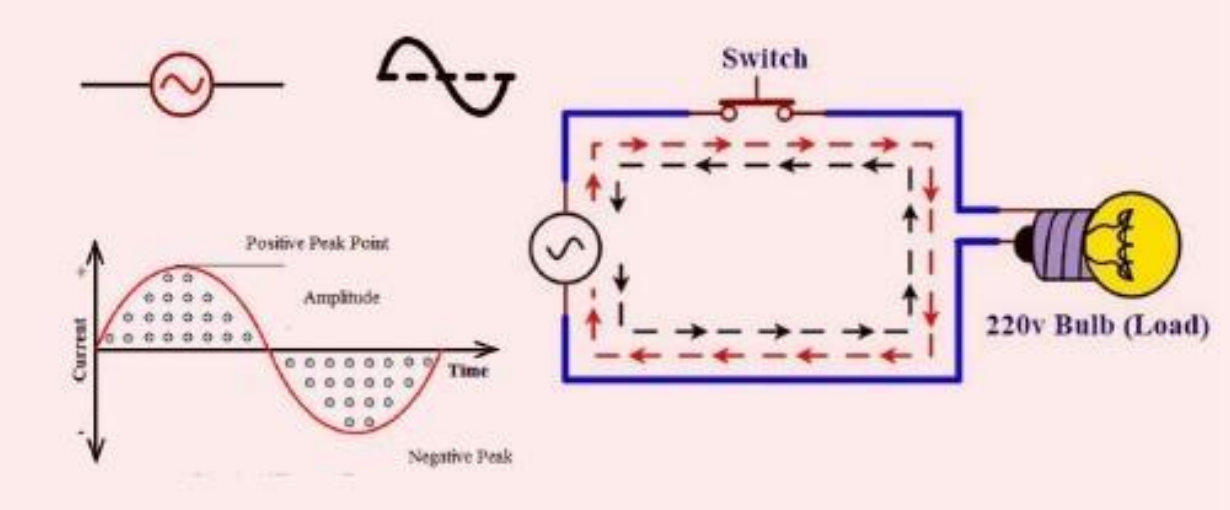
Direct current is defined by the constant flow of electrons from a region of high electron density to a region of low electron density. DC is used in many household appliances and applications that involve a battery.



Alternating Current:

The current electricity that is bidirectional and keeps changing the direction of the charge flow is known as alternating current.

The bidirectionality is caused by a sinusoidally varying current and voltage that reverse directions, creating a periodic back and forth motion for the current. The electrical outlets at our home and industries are supplied with alternating current.



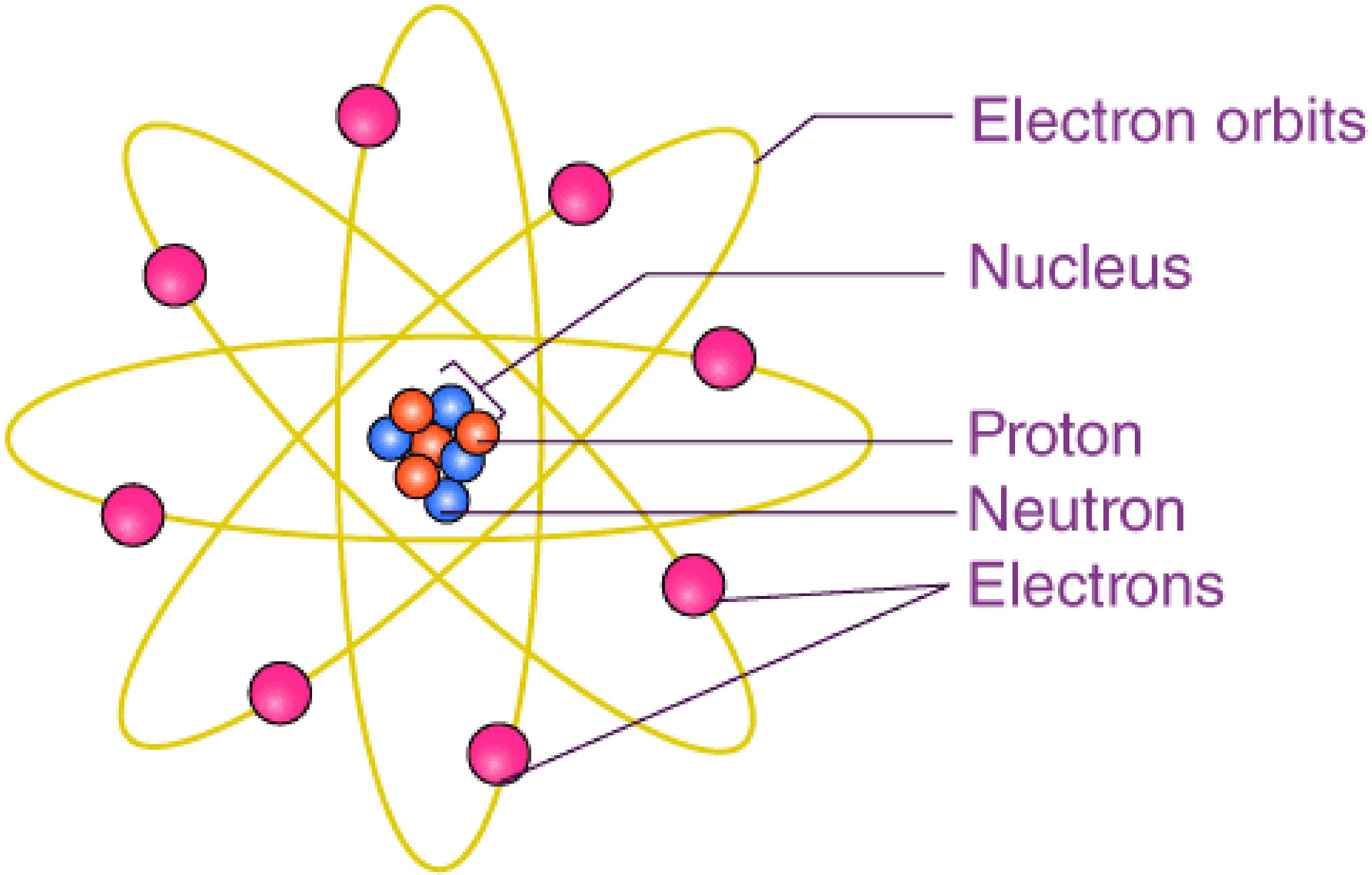
Atoms

Atomic Structure:

Everything in the universe is made of atoms—every star, every tree, every animal. The human body is made of atoms. Atoms are so small that millions of them would fit on the head of a pin.

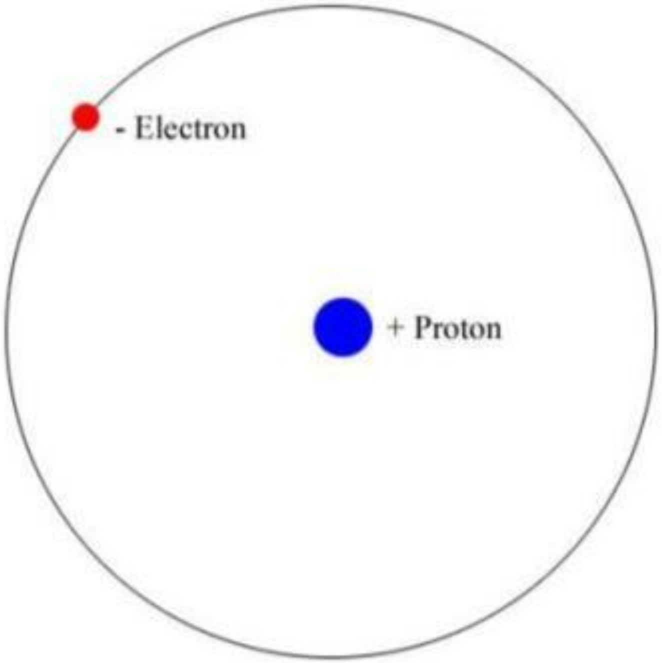
Atoms are made of even smaller particles. **The center of an atom is called The Nucleus**. It is made of particles called **Protons** and **Neutrons**. The protons and neutrons are very small, but electrons are much, much smaller. Electrons spin around the nucleus in shells a great distance from the nucleus.

Atomic Structure:



The Structure of Atom

The Simplest Atom:



- The simplest atom has just one proton and one electron.
- That element is hydrogen.

Nucleus:

In explaining the structure of the atom, the scientist Rutherford said in his atomic model that **a dense, heavy, spherical object is present in a very small space at the center of the atom. He named it the *"Nucleus"***.

About 99.97% of the mass of the atom is concentrated here. All scientists agree that the nucleus is the source of energy in the atom.

The radius of the nucleus is very small (10^{-14} m). Since almost all of the mass of the atom is contained in the nucleus in a very small space, the density of the nucleus is very high (about 10^{17} kg/m^3). **Two types of fundamental particles are seen inside the nucleus- *Proton* and *Neutron*. They are also called *"Nucleons"***.

Exception: Hydrogen atom. Its nucleus contains only protons.

Proton:

- ✓ Its mass is $\mathbf{1.673 \times 10^{-27}}$ **kg** which is close to the mass of the neutron.
- ✓ It is about 1837 times the mass of the electron.
- ✓ The charge of the proton = +e = $\mathbf{+1.6 \times 10^{-19}}$ **coulomb**, which is equal to the charge of the electron but opposite in nature.
- ✓ One proton can be converted into almost **939 MeV** of energy.
- ✓ In 1919, scientist **E. Rutherford** discovered the proton.

Neutron:

- ✓ Its mass is $\mathbf{1.675 \times 10^{-27}kg}$ which is slightly more than the mass of the proton.
- ✓ It is 1839 times the mass of the electron.
- ✓ It is chargeless.
- ✓ If converted to energy, it will release about **942 MeV** of energy.
- ✓ In 1932, scientist **J. Chadwick** (Chadwick) discovered the neutron.

Electrical Units and Symbols

Electrical Parameter	Measuring Unit	Symbol
Voltage	Volt	V or E
Current	Ampere	I or i
Resistance	Ohm	R or Ω
Conductance	Siemen	G or ℧
Capacitance	Farad	C

Electrical Parameter	Measuring Unit	Symbol
Charge	Coulomb	Q
Inductance	Henry	L or H
Power	Watts	W
Impedance	Ohm	Z
Frequency	Hertz	Hz

Unit Name	Unit Symbol	Quantity
Coulomb	C	Electric charge (Q)
Ampere-hour	Ah	Electric charge (Q)
Joule	J	Energy (E)
Kilowatt-hour	kWh	Energy (E)
Electron-volt	eV	Energy (E)
Ohm-meter	$\Omega \cdot m$	Resistivity (ρ)
siemens per meter	S/m	Conductivity (σ)
Volts per meter	V/m	Electric field (E)
Newtons per coulomb	N/C	Electric field (E)

Multiples and Sub-multiples

Prefix	Symbol	Multiplier	Power of Ten
Terra	T	1,000,000,000,000	10^{12}
Giga	G	1,000,000,000	10^{9}
Mega	M	1,000,000	10^{6}
kilo	k	1,000	10^{3}
none	none	1	10^{0}
centi	c	1/100	10^{-2}
milli	m	1/1,000	10^{-3}
micro	μ	1/1,000,000	10^{-6}
nano	n	1/1,000,000,000	10^{-9}
pico	p	1/1,000,000,000,000	10^{-12}

Electrical and Electronic Devices

Electrical Device:

An electrical device is a type of apparatus that uses electricity to perform a specific function. These devices can convert electrical energy into other forms of energy, such as mechanical, thermal, or light energy.

There are two types of electrical devices :

- Active element: Active elements are components that can supply energy to a circuit.
- Passive element: Passive elements are components that can only receive and consume or store energy.

Summarized Key Differences:

Feature	Active Elements	Passive Elements
Energy Supply	Can supply energy	Cannot supply energy
Control	Can control current flow and amplify signals	Cannot control current flow significantly
Power Gain	Can provide power gain or amplification	Cannot provide power gain
Examples	Transistors, diodes, voltage/current sources	Resistors, capacitors, inductors

Electronics:

Electronics can be defined as a branch of physics and engineering that deals with the study of electron behavior, flow, and control under different conditions. Such conditions include vacuums, gases, or semiconducting materials where electrons are observed.

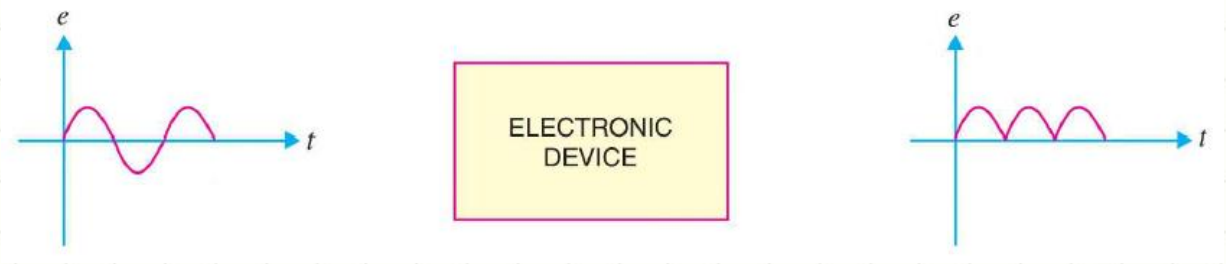
Electronic Devices:

The device which controls the flow of electrons through vacuum or gas or semiconductor for performing a particular task is known as the electronics devices.

Importance of Electronics:

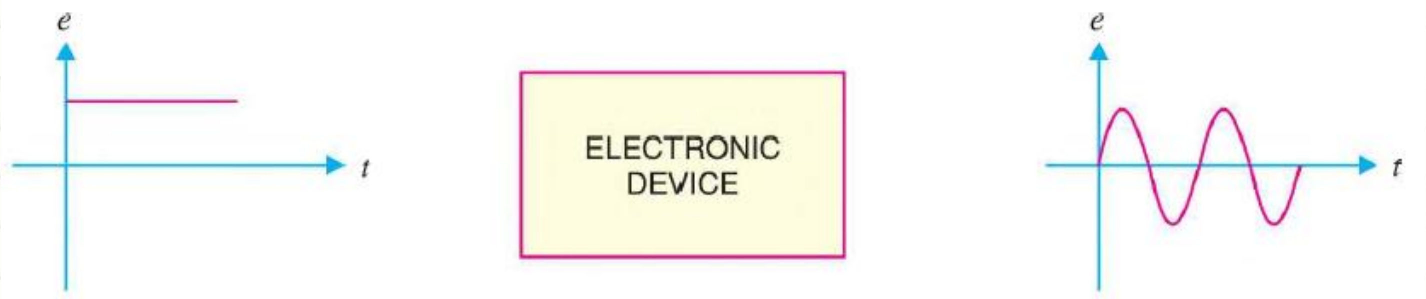
➢ Rectification:

The conversion of AC into DC is called ***Rectification***. Electronic devices can convert AC power into DC power with very high efficiency, they are known as **Rectifiers**.



➢ Generation:

Electronic devices can convert DC power into AC power of any frequency. When performing the action, they are called **Oscillators**.



Importance of Electronics:

➢ Amplification:

The process of raising the strength of a weak signal is called ***Amplification***. Electronic devices can do the job and thus known as amplifiers. For example: An amplifier used in a radio set amplifies weak radio signal so that it can be heard loudly.

➢ Control:

Electronic devices find wide applications in automatic controls. For example: speed of a motor, voltage across a refrigerator etc. can be automatically controlled by using these devices.

➢ Conversion:

Electronic devices can convert light into electricity and vice versa. This conversion of light into electricity is known as ***Photo-electricity***. These devices are used in sound recording on motion pictures. Some can also convert electricity into light. This valuable property is used in television and radar.

QUESTIONS?

See You In The Next Class